Units matter: cooperation in large language model agents tracks how the payoff matrix is written, not what it means

Author One[1] and Author Two[1,2]

[1]Affiliation One
[2]Affiliation Two

Abstract

Multiplying every payoff by a positive constant leaves the outcome ordering, the dominant action, the equilibrium set and the replicator dynamics untouched, so for any agent playing the game the predicted effect is zero by theorem rather than by convention. We apply it to six frontier language models in the iterated prisoner's dilemma, three of them across ten payoff scales spanning five orders of magnitude, over five prompt languages, 15,600 agent-games and 156,000 decisions. Cooperation moves by up to 0.24, far above a measured test-retest floor of 0.04, and it moves in a different shape in each model: oscillating in one, rising and saturating in a second, drifting smoothly downward in a third. Because the shapes differ, pooling across models cancels them and reports a large effect as a small one. On the three-scale grid that most designs sample, a description built from how the numbers are printed beats one saturated in the scale by 8.1 BIC points; we show that this equals the penalty for one parameter at zero difference in fit, and that it reverses on ten scales in every model. Holding the scale fixed and rendering the same payoffs as decimals shifts cooperation by 0.06 in one model against a placebo rendering of 0.02, but by a constant offset that cannot produce dependence on the scale, which survives holding the notation fixed. In a stag hunt the pattern reverses. A cooperation rate reported for a language model is a statement about one arbitrary choice of units, and should be published with its scale sensitivity.

Keywords: machine behaviour; large language models; iterated prisoner's dilemma; payoff scaling; utility invariance; benchmark validity

1 Introduction

Language models are now deployed as *agents* that negotiate, coordinate and decide on behalf of their users [1, 2], and a fast-growing literature measures how they behave when they do. The instrument of choice is the classical game, and the prisoner's dilemma above all, because its strategy space is mapped and its predictions are sharp [3–7]. Studies in this vein report cooperation rates, departures from Nash play, model-specific signatures and archetypes recovered from long games [8–12], and the numbers they produce increasingly feed models of mixed human-machine populations [13–15].

Every one of those measurements is made at a particular payoff matrix. The matrix is written down once, in whatever units the experimenter chose, and the cooperation rate is then reported as a property of the model. The choice of units is not usually stated as a design decision at all, because in the theory it is not one. A payoff matrix is a von Neumann–Morgenstern utility, and such a utility is defined only up to a positive affine transformation [16, 17]. Writing the same dilemma with cells ten times larger changes no preference, no best reply, no equilibrium and no replicator trajectory. Under the theory the two matrices are not two games written differently. They are one game.

That makes a positive rescaling the sharpest possible probe of whether a language model is playing the game it was shown. Most manipulations one might apply to a game-theoretic prompt have arguable effects: a change of framing carries new words, a change of persona carries new information about the agent, a change of language carries a different training distribution. A rescaling carries none of these. It changes the numerals and nothing else. If behaviour is a function of the game, the effect is exactly zero, and the null needs no defending because it is a theorem. If behaviour moves, then whatever the model is responding to is not the game.

This paper takes the rescaling itself as its object of study. We apply it to six frontier models in the iterated prisoner's dilemma, three of them across ten payoff scales spanning five orders of magnitude, crossed with five prompt languages and two persona instructions, and ask four questions.

1. Does the strategic behaviour of a language model survive multiplying every payoff by a positive constant, and by how much does it fail to?

2. If it moves, what does it move with? A rescaling changes the magnitude of the numbers and the way they are printed at the same time. These are different functions of λ, and the data can tell them apart.

3. Where in the game does the effect enter: at the opening move, before any history exists, or through the course of play?

4. What else does the scale carry with it, and what does it leave alone?

Our contributions are the following. **(i)** We report response curves over ten payoff scales from 10^{-2} to 10^{3}, and no two of the three models dense enough to show a shape have the same one: cooperation oscillates in one, rises and saturates in a second and drifts downward in a third, so pooling across models cancels them and reports a large effect as a small one (§3.1). **(ii)** We show that the description built from how the cells are *printed*, which wins on the three-scale grid most designs sample, wins there by exactly the penalty the Bayesian information criterion charges for one parameter at zero difference in fit, and loses in every model once ten scales are available (§3.2). **(iii)** We hold the scale fixed and change only the rendering, and find that a rendering change moves cooperation in one model of two but by a constant offset, which cannot generate a dependence on the scale (§3.3); in a stag hunt the same manipulation reverses which feature matters (§3.4). **(iv)** We show that the effect is already open on the opening move, before any history exists (§3.6), and that of six measures of the same games the rate at which the two agents fail to match is the one that moves least (§3.7). **(v)** We show that the payoff scale shifts how far a persona instruction is obeyed, in the same direction in every model, while leaving the sign of that effect a property of the model; and that the scale interacts with the prompt language (§3.8–§3.9). **(vi)** We show that the strategy a read-out assigns to a run of play moves with the scale in all six models, that the choice of λ reverses the cooperation ranking of two models, and propose a reporting standard that costs one extra condition (§3.10, §3.5, §4.7).

An earlier preprint by the present authors reported that the cooperation rate of open-weight models varies with the payoff scale and with the prompt language [18]. That report established the phenomenon on a set of small open-weight models, which are not the models anyone deploys; the present corpus is entirely frontier, and none of it is shared with that report. The present paper asks what the phenomenon is a function of, and every contribution numbered above is new to it: the frontier replication, the magnitude-against-notation comparison, the localisation of the effect at the opening move, the persona result, the coordination invariant, the strategy read-out, the ranking reversals and the reporting standard. The corpus is the same, re-analysed under the payoff polarity fixed in §2.2.

2 Material and methods

2.1 The invariance under test

Let u be a von Neumann–Morgenstern utility over the four outcomes of a two-by-two game. The representation theorem says that u and $a + b\,u$ with $b > 0$ describe the same preferences [16, 17]. Every quantity a game theorist computes from the matrix is therefore invariant under that map: the ordering of the outcomes, the best-reply correspondence, the set of Nash equilibria, the dilemma indices greed $(T - R)/(T - S)$ and fear $(P - S)/(T - S)$, and the replicator flow, which is invariant under b up to a rescaling of time and under a exactly.

We manipulate the multiplicative half of that map, $u \mapsto \lambda u$ with $\lambda > 0$, and we say so explicitly because the two halves are not equally demanding. The additive half moves every cell by the same amount and can be absorbed into a change of reference point; the multiplicative half changes the number of digits in every cell and is therefore the half that a system reading the matrix as text is most likely to notice. We do not test the additive half here, and we note in §4.6 what that leaves open.

Figure 1a states the stage game and the list of quantities that λ cannot touch. Figure 1b states what it does touch, which is the string the prompt prints.

2.2 The game, the models and the sweep

FAIRGAME (Framework for AI Agents Bias Recognition using Game Theory) [19] provides the computational infrastructure. Conditions are written in JSON; at run time the framework combines them with language-specific prompt templates, simulates repeated normal-form games and logs every round of play. We adopt the framework and its multilingual templates as given, which keeps the wording of the manipulations out of our hands, and we add the payoff scaling.

Agents played a symmetric two-player prisoner's dilemma. The templates state the cell values as *penalties*, that is, years of imprisonment in the prisoner's-dilemma story, and instruct the agent to *minimise* them, so a smaller number is a better outcome and the ordering is $T < R < P < S$. That is the penalty-space image of the usual $T > R > P > S$, and it makes Option A the dominant, defecting action and Option B cooperation. We state the direction explicitly because reversing it would reverse the sign of every result below. The assignment is not a matter of interpretation: FAIRGAME logs the realised payoff next to every action, and the assignment above reproduces the logged payoff on all 264,000 rounds of the corpus while the opposite assignment reproduces none of them.

The corpus is a single arm of frontier models playing a single game (figure 1c). Six models met the same base matrix, the same five prompt languages and the same four ordered persona pairings, over ten rounds. They differ only in how densely they sample the scale.

Three models carry the resolution that the question of shape requires. Gemini 3.5 Flash-Lite, Gemini 3.1 Flash-Lite Preview and GPT-5.4 Nano each met ten payoff scales, $\lambda \in \{0.01, 0.1, 0.25, 0.5, 1, 2, 5, 10, 100, 1000\}$, spanning five orders of magnitude. The grid is deliberately not a pure decade grid. The four off-decade scales $\{0.25, 0.5, 2, 5\}$ are what allow magnitude and printed form to be told apart at all, because $\lambda = 0.5$ prints as integers while $\lambda = 0.25$ prints with decimal points and the two are a factor of two apart; on a decade grid no such pair exists.

Three further models, Claude 3.5 Haiku, GPT-4o and Mistral Large, met three scales, $\lambda \in \{0.1, 1, 10\}$. These are the models our inference infrastructure could not extend, and we use them only for results that do not require resolution in λ. They contribute nothing to the description comparison of §3.2, and we say so there.

Every cell is balanced at exactly 40 dyads and 80 agent-games in every model $\times$ language $\times$ scale combination, with none missing, which gives 7,800 dyads, 15,600 agent-games and 156,000 binary decisions. Two further manipulations, described in §3.3 and §3.4, add 2,400 and 1,600 dyads on top of that.

The base matrix is $T = 0$, $R = 2$, $P = 6$, $S = 10$, which is greed 0.20 and fear 0.40 on the utility scale $u = -$penalty, and it is constant across every model and every scale. The corpus is therefore one game shown at several scales, which is the comparison every result below is made within.

On each round an agent received the full payoff table, its persona, the round index and the complete history of both players' choices, and returned a single token naming one of two options. Communication was disabled, so the only channel between agents is the action history. Actions carry the labels `OptionA` and `OptionB` in all five languages, which avoids the extra meaning that "cooperate" and "defect" would bring; the templates never say which label means cooperation, and the agent has to read it off the payoff table. That is what makes the payoff numbers the load-bearing part of the prompt, and it is why a change to them is the manipulation of interest.

Decoding used a fixed low output-token budget with reasoning traces disabled. That setting is not a detail. A model that emits hidden reasoning can exhaust a small budget inside the trace and return an incomplete string, and a harness that then falls back to a default action turns that string into an apparent cooperation rate of 1.0 that the parser has created rather than the model. We logged parse-failure and fallback rates for every cell and required both to stay below 0.02. Every dyad ran its full horizon and every decision resolved to one of the two option tokens, so no cell loses games to a failure to parse.

2.3 Notation regimes

Because λ changes the magnitude of the payoffs and the way they are printed at the same time, we define the printed quantities in advance and let the model comparison of §3.2 arbitrate between them. For each scale we take the four cell values as the template renders them and record three features: whether any of them carries a decimal point, the mean number of glyphs across the four, and a three-level *notation regime* with levels *fractional* (at least one cell printed with a decimal point), *unit* (all cells integers, largest three digits or fewer) and *large* (all cells integers, largest four digits or more). Figure 1b lists the rows.

The regime is read off the printed cells and not off λ, and on this grid the distinction is not cosmetic. On a decade grid the two readings coincide exactly, which is why earlier work could not tell them apart; here they part company at $\lambda = 0.5$, whose cells print 0/1/3/5 and are therefore *unit* even though $\lambda < 1$. We report both operationalisations in §3.2 and give the one read from the printed string priority, because it is the one the agent actually sees.

Even so, the grid alone cannot finish the argument. Across it, magnitude and printed form remain correlated: every scale that prints with decimal points is also a small-stakes scale, give or take the single off-decade exception above. No arrangement of λ removes that, because the T cell is zero at every scale and therefore prints as `0` under integer rendering and `0.00` under decimal rendering, so the two accounts can never be fully crossed by choosing scales. Separating them requires holding λ fixed and changing only the rendering, which is a second experiment rather than a different analysis of this one. We report it in §3.3.

2.4 The strategy read-out, and where its labels come from

A cooperation rate averages over a game. The second summary the literature reports, and the one that feeds evolutionary models, is the strategy assigned to a whole run of play, so we measure that too. Because the assignment is made by an instrument rather than observed, the instrument is described here in full and its output is reported by provenance rather than as a single label column.

The vocabulary is the four canonical memory-one rules: AllC, AllD, tit-for-tat [3] and win-stay lose-shift [20]. The read-out runs in two stages. Every one of the four prescribes an action as a function of the previous round's joint outcome alone, so the first stage checks by enumeration whether a trajectory is *exactly* what a rule would have played, and returns the set of rules that fit; no learning is involved and the answer is a proof. The first stage has three outcomes: exactly one rule fits, and the label is provable; several fit, in which case the run is recorded as *ambiguous*, which happens whenever the realised history never discriminates them, as when an agent cooperates throughout against a cooperator and is thereby AllC, TFT and WSLS at once; or none fits. Where none fits, a second stage supplies the nearest rule: a single-layer long short-term memory network [21] with a 24-dimensional embedding and 96 hidden units, trained on synthetic play generated from the same four rules at zero and five per cent execution noise, returns a posterior over them and we take its argmax. Every game in this corpus runs ten rounds, so a single network trained at that horizon reads all of it.

Two properties of that pipeline bear on how its output should be read, and both are reported rather than assumed away. The first is provenance. Over the whole corpus 19.4% of agent-games carry a provable rule, 16.4% are ambiguous and 64.3% are named by the network. The split differs between the two groups of models, at 17.2%, 20.1% and 62.7% for the three that met ten scales against 26.7%, 3.9% and 69.4% for the three that met three. Two thirds of the labels are therefore inferences and not deductions, and a further 24.4% of all agent-games sit below a posterior of 0.90, a floor at which a read-out designed to abstain would decline to answer at all. The second is validation. On held-out synthetic play at the ten-round horizon this corpus uses, the pipeline recovers the generating rule with accuracy 0.975 against a Bayes-optimal single-label ceiling of 0.987, and where a rule fires uniquely the label is right 99.5% of the time. Frontier transcripts depart from the canonical rules by more than the training distribution does, so their labels are nearest neighbours more often than they are recognitions. The electronic supplementary material, §S1, gives the full validation, including how the network degrades on noise levels it never saw.

We therefore treat the strategy mix as a second, coarser dependent variable rather than as ground truth, and we report every test on it twice: once over the five categories the read-out emits, and once over the four named rules alone. The distinction matters because *ambiguous* is a statement about how identifiable the play is and not about which strategy was played, so a shift into or out of that category is a shift in nameability. Where the two versions of a test disagree we say so.

2.5 Statistical analysis

All intervals are 95% bootstrap confidence intervals over 2,000 resamples that resample whole dyads [22]. The choice of unit matters: rounds within a game are autocorrelated and the two agent-rows of a dyad share a history, so a row-level resample would narrow the interval by roughly a factor of $\sqrt{2}$.

Whether a factor moves a measure at all is answered against a permutation null that shuffles the factor label across *dyads*, so a dyad keeps both of its agent-rows together and the null preserves the dependence the design creates. We report the observed largest between-level gap, the permutation P value over 2,000 draws, and the null's 95th percentile, which doubles as the smallest shift the test could have detected. For distributions over strategy labels we use the largest total-variation distance between two levels, with the same dyad-level null.

Whether the response is a function of the magnitude or of the notation is a model-comparison question, and we answer it with the Bayesian information criterion [23] over a ladder of specifications fitted to identical rows. Every specification carries the same controls for model, language, own persona and opponent persona, and they differ only in how λ enters: not at all; linearly, quadratically or cubically in $\log_{10} \lambda$; as the fractional flag; as the fractional flag plus the glyph count; as the three-level notation regime; or saturated, as a free level per scale. The saturated specification is the upper bound on fit that any function of λ can reach, so a description that matches it at fewer parameters is a description that has found the right shape.

We do not correct for multiple comparisons across models, and we say so rather than leave the point to be inferred. Each model is a separate system and is its own family, so the tests within a manipulation are six families of one rather than one family of six. The claims we draw are in any case about whether the rescaling moves behaviour at all, not about which model moves most.

2.6 Confounds we declare

Two features of the design vary between models, and one affects a single cell. Models differ in whether the round count was disclosed in the prompt, which it was for the three models with the full sweep and was not for the three with three scales, and in sampling temperature, which follows each provider's recommended default as in the FAIRGAME protocol. Both are constant within a model, so neither can generate a within-model effect of λ, which is what every result below is. They do mean that no level statistic is comparable between the two groups of models, and we compare none: every claim about the shape of the response is made within a model.

A third feature is confounded with the comparison of §3.3 rather than with λ, and we treat

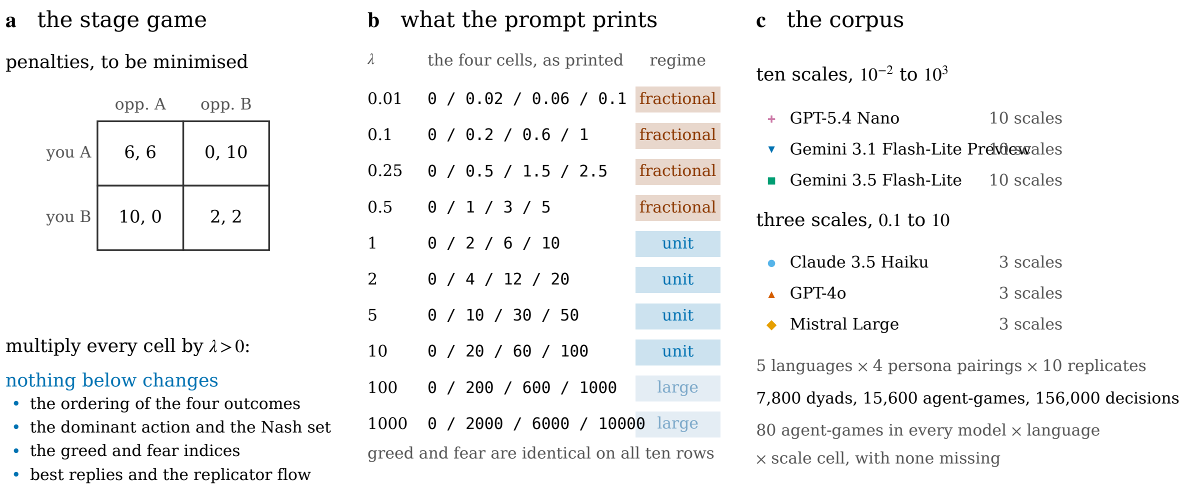

Figure 1: The game, the rescaling and the corpus. (**a**) The stage game at $\lambda = 1$. Cells are penalties the agent is told to minimise, so $T < R < P < S$ and Option A is the dominant, defecting action. Multiplying every cell by $\lambda > 0$ is the multiplicative half of a positive affine transformation of a von Neumann–Morgenstern utility, and leaves every listed quantity where it was. One matrix is used throughout the corpus. (**b**) What the rescaling does change: the four cell values as the prompt prints them, at each of the ten scales, with the three-level notation regime of §2.3 on the right. Greed and fear are identical on every row. The row for $\lambda = 0.5$ is the one that separates the two readings of the regime: it prints as integers although $\lambda < 1$. (**c**) The corpus. Every cell holds 80 agent-games in every model × language × scale combination. Three models sample ten scales and three sample three.

it there: the rendering conditions were collected as separate runs, so a run-level offset is not separable from the rendering by design alone. The placebo condition of that experiment is what measures the offset, and we report it alongside the effect.

Separately, four dyads in the Claude × English × $\lambda = 10$ cell are logged with the payoffs of $\lambda = 1$. We recovered the realised scale of every game from the logged values rather than assuming it from the directory name, which is how we found them; they are 8 agent-games out of 12,000, and dropping them moves Claude's cooperation rate at $\lambda = 10$ from 0.708 to 0.715. We retain them at their nominal scale and note that no conclusion turns on them.

3 Results

3.1 A rescaling that cannot matter, and does

Multiplying every payoff by a positive constant moves the cooperation rate (figure 2). Per model the largest shift between two scales is 0.243 for GPT-5.4 Nano, 0.240 for Claude 3.5 Haiku, 0.234 for GPT-4o, 0.191 for Gemini 3.5 Flash-Lite, 0.072 for Gemini 3.1 Flash-Lite Preview and 0.036 for Mistral Large. Four of the six exceed a permutation null that shuffles the scale label across whole dyads, at $P < 0.001$ against null 95th percentiles between 0.058 and 0.082 (table 3). The same game, written with a different number of digits, is played differently.

Before reading any of those numbers we fix the floor beneath them. We reran one model at four scales with a fresh seed, holding everything else identical, and compared the two runs cell by cell: the median absolute difference is 0.008 and the largest is 0.038, over 800 fresh dyads. Nothing below roughly 0.04 is interpreted anywhere in this paper, and every effect we do interpret is stated against that floor.

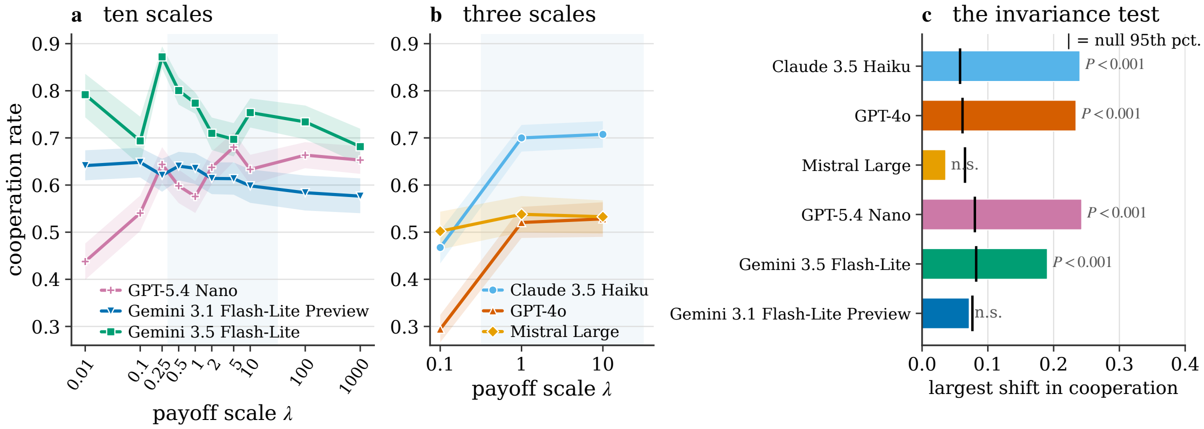

Figure 2: Cooperation against the payoff scale. (**a**) The three models that met ten scales. Bands are 95% bootstrap intervals over 2,000 resamples of whole dyads. The shaded strip marks the unit regime. No two of the three curves have the same shape: one oscillates, one rises and saturates, one drifts smoothly downward. (**b**) The three models that met three scales, on the same axes. The window they sample is too narrow to show any of those shapes, and it suggests a different one for each model. (**c**) The invariance test. Bars give the largest shift in cooperation between two scales; ticks give the 95th percentile of a permutation null that shuffles the scale label across whole dyads. The shift a theory of the game predicts is exactly zero. Four of the six models exceed their null; the two that do not are discussed in §3.1, and they fail it for different reasons.

Two models do not clear the permutation null, and they fail it for different reasons, which is worth separating rather than reporting as one exception. Mistral Large moves 0.036, $P = 0.38$, and that is smaller than the noise floor: on this measure its behaviour is invariant to the rescaling in the way the theory says it should be. Its strategy mix is not, and we return to that in §3.10. Gemini 3.1 Flash-Lite Preview moves 0.072, $P = 0.094$, which is above the floor but not beyond a null built for jumps. Its response is a slow monotone drift across ten scales rather than a jump between any two of them, and a statistic defined as the largest gap between two cells has almost no power against that shape. The description comparison of §3.2, which uses all ten cells at once, puts the same model's dependence on λ at 37.1 BIC points over a specification with no λ term at all. The two tests disagree because they are powered for different alternatives, and we report both rather than the one that flatters the claim.

The shape of the response is the first substantive result, and it is only visible because three models sweep five orders of magnitude at ten scales. It is not monotone, it is not the same shape in different models, and the second of those is the finding that matters.

Gemini 3.5 Flash-Lite oscillates. Cooperation runs 0.792, 0.694, 0.872, 0.800, 0.774, 0.710, 0.697, 0.754, 0.734, 0.681 across the sweep, changing direction five times, with five adjacent steps individually beyond the noise floor and opposite in sign. GPT-5.4 Nano rises and then saturates: 0.438, 0.540, 0.644, 0.598, 0.576, 0.637, 0.680, 0.633, 0.663, 0.653, with the two steps out of the smallest scales worth 0.103 each ($P < 0.001$) and nothing beyond $\lambda = 5$. Gemini 3.1 Flash-Lite Preview drifts smoothly downward, 0.648 to 0.577 over five orders of magnitude, with no single adjacent step reaching significance (smallest $P = 0.25$).

Because those three shapes are not the same shape, they partly cancel. Pooled over the three models the curve is nearly flat, and a specification with no λ term at all sits only 8.4 BIC points behind the best one, against 80.9, 37.1 and 220.9 points in the same three models taken

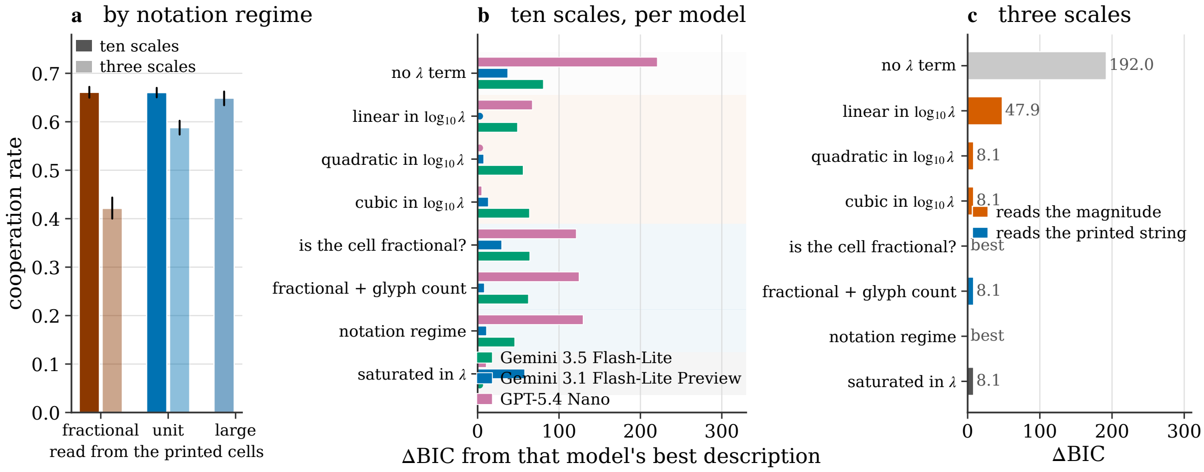

Figure 3: What the response is a function of. (**a**) Cooperation by notation regime, read from the printed cells. Whiskers are 95% dyad bootstrap intervals. The three regimes are within a point of each other once the sweep is dense enough to separate the regime from the scale. (**b**) The description ladder, one bar per model, for the three models that met ten scales. Bars give the BIC penalty each specification pays relative to the best one for that model, with language and both personas controlled in every row; the background bands mark the two families of description. The best specification differs between the three models, and no notation specification is the best one for any of them. (**c**) The same ladder computed on the three-scale models, pooled as those models were designed to be pooled. The notation specifications win there by 8.1 points over the specification saturated in λ, which is the penalty for one parameter at a difference in r^2 of 0.00002: a three-point grid cannot tell these specifications apart, and four of the eight rungs are numerically the same model.

separately (§3.2). Averaging models is therefore not a conservative choice here; it is a way of reporting a large effect as a small one, and we avoid it throughout.

One consequence is immediate for how such experiments are designed. A three-scale window at $\{0.1, 1, 10\}$, which is what most published work samples, would report a rising response in GPT-5.4 Nano ($0.540 \to 0.576 \to 0.633$), a falling one in Gemini 3.1 Flash-Lite Preview ($0.648 \to 0.635 \to 0.598$) and a rise-then-fall in Gemini 3.5 Flash-Lite ($0.694 \to 0.774 \to 0.754$). The narrow window is not merely imprecise about the shape. It reports a different shape for each model, and none of the three is the shape the wider sweep shows.

3.2 Magnitude or notation

On three scales, the notation account wins. Restricted to the three models that met $\lambda \in \{0.1, 1, 10\}$, and pooled as those models were designed to be pooled, the fractional flag is the best specification in the ladder: it beats a specification with no λ term by 191.9 BIC points, a linear term in $\log_{10}\lambda$ by 47.9, and the specification saturated in λ by 8.1 (table 4). That is the result the narrow design supports, and it is the result we would have reported had we stopped there.

It does not survive contact with its own arithmetic. On three scales the saturated specification has two free λ parameters and the fractional flag has one, and the two fit the data equally well: $r^2 = 0.147946$ against 0.147930. A BIC difference at equal fit is the parameter penalty and nothing else, and the penalty for one parameter at $n = 3{,}600$ agent-games is $\ln n = 8.19$. The observed advantage is 8.12. The notation account did not describe the data better; it described the data identically with one parameter fewer, on a grid too coarse for the distinction to mean anything. On three points, *quadratic*, *cubic*, *fractional flag plus glyph count* and *saturated* are numerically the same model, with BIC equal to six decimal places, because three points cannot tell them apart.

On ten scales the conclusion reverses, in every model that has ten scales (figure 3b, table 4). Against the best specification for each model, the three-level notation regime pays 45.7 BIC points in Gemini 3.5 Flash-Lite, 11.0 in Gemini 3.1 Flash-Lite Preview and 129.7 in GPT-5.4 Nano. No notation specification wins anywhere. What wins is different in each model, and the differences are large: the specification saturated in λ in Gemini 3.5 Flash-Lite, ahead of the notation regime by 45.7; a linear term in $\log_{10} \lambda$ in Gemini 3.1 Flash-Lite Preview, with the saturated specification last of all, 57.9 points behind and worse than having no λ term; and a quadratic in $\log_{10} \lambda$ in GPT-5.4 Nano, which is the shape of a curve that rises and levels off, ahead of the linear term by 67.3.

One detail decides between the two ways of reading a notation regime, and it points the same way. The regime can be defined from λ, as earlier work necessarily did on a decade grid, or read off the printed cells. The two differ at exactly one scale in this design, $\lambda = 0.5$, whose cells print 0/1/3/5 and are therefore integers despite $\lambda < 1$. The reading from the printed cells is the one the agent can act on, and it is the worse of the two in both models where the comparison is possible: 62.0 against 45.7 in Gemini 3.5 Flash-Lite, 15.7 against 11.0 in Gemini 3.1 Flash-Lite Preview. Whatever advantage a notation description had on the decade grid came from its coinciding with a function of λ there, and not from how the numbers look.

The pairwise steps say the same thing without any model selection. In all three models the largest step between adjacent scales crosses no notation boundary: +0.178 from $\lambda = 0.1$ to 0.25 in Gemini 3.5 Flash-Lite and +0.103 from 0.01 to 0.1 and again from 0.1 to 0.25 in GPT-5.4 Nano, all of them inside the fractional regime. The one step in the sweep that does cross the boundary, $\lambda = 0.25$ to 0.5, is smaller in every model: −0.071, +0.020 and −0.046.

The three models with three scales are silent on all of this, and we say so rather than pooling them in. With three points their ladder is degenerate in the way shown above, so they cannot distinguish shapes; they enter the paper only through results that need no resolution in λ.

3.3 Holding the magnitude fixed and changing only the notation

Everything above is observational in the same way: it compares scales, and at every scale the magnitude and the printed form move together. The experiment that separates them holds λ fixed and changes only how the four cells are rendered. We ran it on the two models for which we had the full sweep, at $\lambda \in \{0.1, 1, 10\}$, in three rendering conditions.

The *native* condition is the one used everywhere else in the paper and prints each cell as the template renders it, so at $\lambda = 1$ the row reads 0/2/10/6. The *two-decimal* condition prints every cell to two decimal places, 0.00/2.00/10.00/6.00. The payoffs are identical to the last bit; only the string differs, and every integer cell has acquired a decimal point. The comparison between them is the notation manipulation.

The third condition is the one that makes the second interpretable. A shift between two conditions collected as separate runs is not by itself evidence about notation, because separate runs differ. The *three-decimal* condition prints 0.000/2.000/10.000/6.000: every cell is already a decimal in both conditions and the only difference is one further zero, so no notation boundary is crossed at all. Whatever this contrast produces is what a change of rendering produces when the rendering change carries no information, and it is the yardstick for the contrast that does.

We note the design constraint that makes this third condition necessary rather than merely prudent. The T cell of the matrix is zero at every scale, so it prints as `0` under integer rendering and `0.00` under decimal rendering whatever λ is. No choice of scales can therefore produce a pair of cells that differ in magnitude but not in whether any cell carries a decimal point, which is why the placebo has to live on the rendering axis and not on the λ axis.

The result differs between the two models, and only one of them supports a notation effect. In Gemini 3.1 Flash-Lite Preview the two-decimal rendering raises cooperation at all three scales, by 0.046 ($P = 0.029$), 0.070 ($P = 0.002$) and 0.065 ($P = 0.005$), mean absolute shift 0.060, while the placebo contrast gives -0.027, $+0.008$ and -0.028, none of them significant, mean absolute shift 0.021. The effect is three times the yardstick, it has the same sign at every scale, and it is above the noise floor of §3.1 where the placebo is not. For that model, printing the same payoffs as decimals makes the agent about six points more cooperative.

In Gemini 3.5 Flash-Lite it does not. The same contrast gives -0.059, -0.047 and $+0.051$: it changes sign between $\lambda = 1$ and $\lambda = 10$, and the placebo contrast contains a shift of -0.058 at $P = 0.005$, larger than two of the three effects it is supposed to calibrate. A mean absolute shift of 0.052 looks comparable to the other model's 0.060, and reporting only that number would make the two look alike; the sign pattern is what separates them, and an effect that reverses direction between two scales is the shape of noise rather than of a boundary.

The other half of the design is the one that matters most for §3.1. Within a single rendering condition, every cell at every scale prints with the same kind of glyph, so any remaining movement across λ cannot be attributed to cells changing from integers to decimals. It does not go away. Under two-decimal rendering, Gemini 3.5 Flash-Lite moves $+0.092$ from $\lambda = 0.1$ to 1 ($P = 0.004$) and $+0.078$ from 1 to 10 ($P < 0.001$); under three-decimal rendering Gemini 3.1 Flash-Lite Preview moves $+0.046$ ($P = 0.024$) and -0.078 ($P < 0.001$) across the same two steps. The dependence on the payoff scale survives holding the notation fixed.

Read together, the two halves say something more specific than either the notation account or its denial. A rendering change can move cooperation, in some models, by about the size of the smallest effects we interpret. But what it produces is a roughly constant offset, the same at $\lambda = 0.1$ as at $\lambda = 10$, and a constant cannot be what makes the response depend on λ. That is why a notation description can be causally real in one model and still lose the description comparison in that same model by 11.0 BIC points. Both statements are true at once, and the apparent tension between them dissolves once the effect is seen to be a level shift rather than a shape.

3.4 The same manipulation in a second game

A result about one game may be a result about that game. We repeated the sweep in a stag hunt, at $\lambda \in \{0.1, 0.5, 1, 10\}$ on the same two models, 1,600 dyads in all. The game is a coordination problem rather than a dilemma: mutual stag-hunting is the payoff-dominant equilibrium and mutual hare-hunting the risk-dominant one, so there is no dominant action and the question the agent faces is what the other will do, not whether to exploit them. The templates are the FAIRGAME stag-hunt templates, rendered in the same five languages, and the frame is one of rewards to be maximised rather than penalties to be minimised, which reverses which option counts as cooperation. We report stag-hunting.

The pattern reverses with the game. In the prisoner's dilemma no step across the notation boundary was significant in either model. In the stag hunt the step from $\lambda = 0.1$ to $\lambda = 0.5$, which is where the printed cells stop carrying decimal points, moves stag-hunting by -0.201 ($P < 0.0001$) in Gemini 3.1 Flash-Lite Preview and -0.117 ($P = 0.005$) in Gemini 3.5 Flash-Lite: same sign, comparable size, in two models independently. The step from $\lambda = 0.5$ to $\lambda = 1$, which changes the magnitude by a factor of two and changes no notation, moves nothing in either model ($+0.044$, $P = 0.25$; $+0.055$, $P = 0.15$).

We report one asymmetry rather than smoothing it. Gemini 3.1 Flash-Lite Preview also has a significant step where only the magnitude changes, from $\lambda = 1$ to 10, worth $+0.175$ ($P < 0.0001$). For that model both features move behaviour, and the notation step is the larger of the two despite being the smaller change in magnitude, a factor of five against a factor of ten. In Gemini 3.5 Flash-Lite the same magnitude step moves nothing ($+0.028$, $P = 0.46$).

The conclusion we draw is narrower than either side of the argument as it is usually put. Notation is not a general property of how language models read payoff matrices, and neither is its absence. In the stag hunt it moves both models; in the prisoner's dilemma it moves neither; and whether magnitude matters on top of it depends on the model. What is stable across both games and all models is the thing the invariance forbids, which is that the response depends on the scale at all.

3.5 Which model looks more cooperative depends on the units

The consequence for measurement is direct. Among the three models that met ten scales, the largest swing inside a single model, 0.243 for GPT-5.4 Nano, is larger than the median spread between the three at a fixed scale, 0.154. Which λ the experimenter wrote down therefore matters more, for the number that comes out, than which of the three models was asked. The three models that met three scales sit the other way round, and only just: there the median between-model spread is 0.180 against a largest within-model swing of 0.240.

Rankings reverse. Of the 135 model-pair by scale-pair comparisons available among the ten-scale models, 24 change sign, and of the 9 available among the three-scale models, 2 do. Concretely, GPT-5.4 Nano cooperates 0.204 less than Gemini 3.1 Flash-Lite Preview at $\lambda = 0.01$ (0.438 against 0.641) and 0.080 more at $\lambda = 100$ (0.664 against 0.584), so the sign of a difference that a benchmark reports as a single number is set by the units. Among the three-scale models Mistral Large cooperates more than Claude 3.5 Haiku at $\lambda = 0.1$ (0.502 against 0.468) and less at $\lambda = 1$ (0.538 against 0.700). A benchmark that fixes one scale reports one side of these comparisons and has no way to signal that the other side exists.

3.6 The effect is set at the opening move

A cooperation rate is an average over a game, so a difference between two scales could enter anywhere in it. It enters at the first move (figure 4a,b). Opening cooperation, measured on round one alone, swings by 0.163 across the ten scales ($P < 0.001$, null 95th percentile 0.065) and by 0.182 across the three ($P < 0.001$, null 0.049). In both groups that is larger than the swing in the whole-game cooperation rate computed on the same rows, 0.089 and 0.168. The effect is not accumulated over play. It is fully present before play starts, and play then damps it rather than building it.

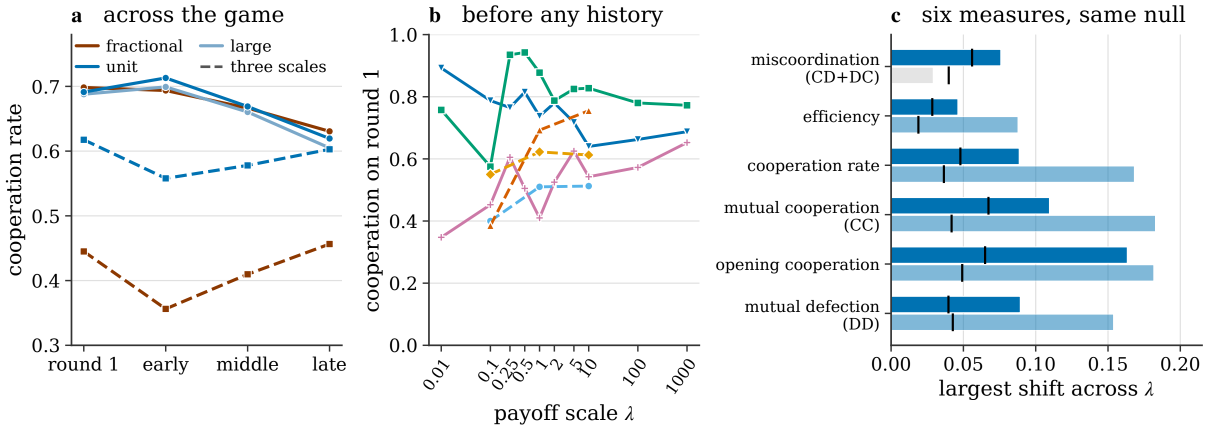

Figure 4: Where the effect enters, and what it moves least. (**a**) Cooperation by notation regime across the game. Solid lines are the three models that met ten scales, dashed the three that met three; the four positions are round one and three equal blocks of the remaining rounds. On the three-scale grid the gap between the regimes is open at round one, before any history exists, and stays open. On the ten-scale grid the three regimes sit within two points of each other at every position, which is figure 3a read round by round. (**b**) Opening-move cooperation against the payoff scale, one line per model, colours as in figure 2. (**c**) The largest shift across scales for six measures of the same games, against the same permutation null. Dark bars are the ten-scale models, pale the three-scale ones; grey marks a measure whose shift does not clear its null. Cooperation and the opening move shift in both groups; the rate at which the two agents fail to match is the smallest mover in both, and the only one of the six that fails its null anywhere.

Round one is the round on which no history exists. The agent has seen the payoff table, its persona and the round index, and nothing else. Whatever the rescaling does, therefore, it does to the prior over actions that the prompt induces, and not to anything the agent learns from its opponent.

Read through the notation regimes rather than through λ, the same games separate the two groups of models, and the difference is worth stating because it is where a coarse grid misleads. On the three-scale grid, where *fractional* is exactly $\lambda = 0.1$ and *unit* is $\lambda \in \{1, 10\}$, the gap between the regimes is -0.173 on round one against -0.167 over the whole game, so the opening move carries all of it and the contrast looks like a notation effect. On the ten-scale grid, where the two readings come apart, the same regime gap is 0.007 on round one and 0.001 over the whole game (electronic supplementary material, table S10). The regime contrast that carries the effect on three scales is not there at all on ten, which is the round-by-round version of §3.2. What remains at the opening move on ten scales is the dependence on λ itself, which is what figure 4b plots.

3.7 What the rescaling moves least

A corpus in which every measure moves is a corpus one should be suspicious of, and on the denser grid this one is close to that (figure 4c). Measured on the same games and against the same dyad-level null, the cooperation rate, mutual cooperation, mutual defection, efficiency and the opening move all move, in both groups of models, at $P < 0.001$. So, across ten scales, does the rate of miscoordination, that is, the share of rounds in which the two agents chose differently: it swings by 0.076 ($P = 0.0005$, null 95th percentile 0.056). We report that rather than the

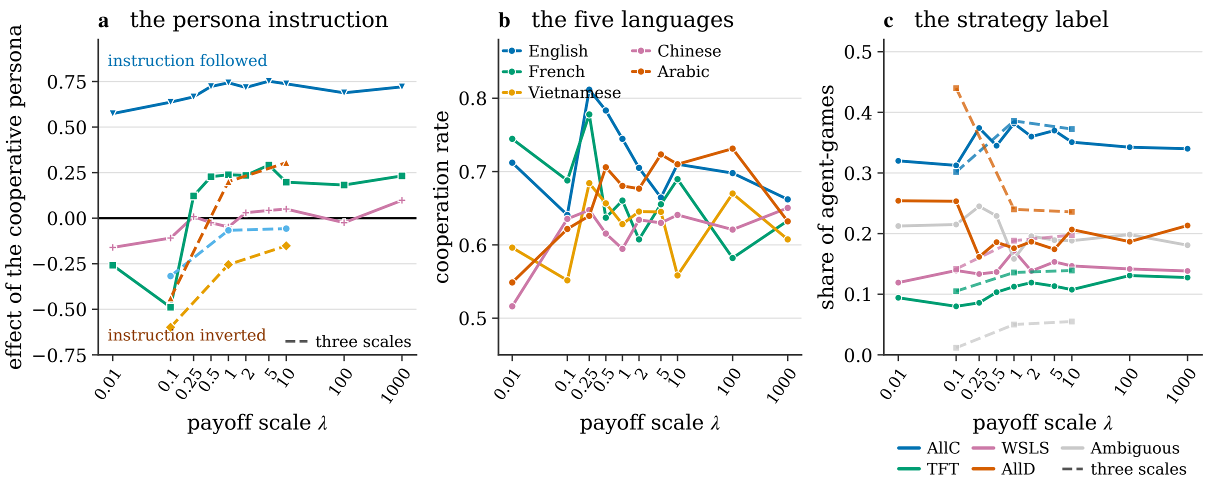

Figure 5: What the payoff scale carries with it. (**a**) The effect of being told to be cooperative rather than selfish, scale by scale, *one line per model*; colours as in figure 2, solid the three models that met ten scales and dashed the three that met three. Every model moves upward across the fractional-to-unit boundary, but only three of the six cross zero: the sign of the effect is a property of the model. The average of these lines crosses zero although half of them never do, which is why no pooled curve is drawn. (**b**) Cooperation by prompt language across the sweep, ten-scale models. The five languages do not move together. (**c**) The share of agent-games carrying each strategy label, against the payoff scale, with the two groups of models encoded as in figure 4a. The two unconditional rules carry most of the movement in both, and much more of it on the coarser grid.

weaker version of the same test, because a coarser grid had this measure as an invariant and ten scales say it is not one.

What survives is an ordering rather than an invariance, and the ordering is the same in both groups. Against its own null, miscoordination is the smallest mover of the six across the ten scales, at 1.35 times the null 95th percentile, against 1.62 for efficiency, 1.85 for the cooperation rate, 2.25 for mutual defection and 2.51 for the opening move. Across the three scales it is the one measure of the six that does not clear its null at all, swinging 0.029 against a null of 0.040 at $P = 0.21$, while mutual cooperation over the same rows swings 0.183. Scaling the payoffs moves how much the agents cooperate a good deal more than it moves how well they coordinate, and on the narrower grid it moves the second not at all. Two agents at $\lambda = 0.1$ and two at $\lambda = 10$ agree with each other about equally often; what differs is what they agree on.

The asymmetry between the two positions in a dyad behaves the same way. Across the ten scales the first-listed agent's cooperation rate differs from the second's by between -0.066 and $+0.026$, with no trend across the sweep; across the three the gap runs -0.002, -0.094 and -0.097.

3.8 The scale shifts how far the persona instruction is obeyed

Each agent received a system-prompt persona, either *cooperative* or *selfish*. A persona is not irrelevant by theorem, but it carries no information about the game: the four pairings present identical payoff matrices, and an agent was never told its opponent's persona. What the persona does carry is an instruction, and how far that instruction is obeyed turns out to depend on the payoff scale (figure 5a, table 1).

The result has to be stated at the level of the individual model, because the pooled average is misleading here in a way worth setting out. Pooled over the three models that met ten scales, the effect of the cooperative persona is +0.052 [0.006, 0.098] at $\lambda = 0.01$, +0.311 [0.275, 0.346] at $\lambda = 1$ and +0.350 [0.314, 0.386] at 1000; pooled over the three that met three scales it runs −0.452 [−0.486, −0.416], −0.040 [−0.081, 0.002] and +0.032 [−0.009, 0.074]. Read alone, those two curves say that the instruction is inverted or inert at the small scales and obeyed above them. That reading does not survive disaggregation. In three of the six models the effect keeps one sign across the entire sweep: positive at every scale for Gemini 3.1 Flash-Lite Preview, negative at every scale for Claude 3.5 Haiku and Mistral Large, and every one of those estimates excludes zero (table 1). The other three, GPT-5.4 Nano, Gemini 3.5 Flash-Lite and GPT-4o, cross zero within the sweep. The pooled curve is therefore an average over models that reverse and models that do not, and nothing in it distinguishes the two: Gemini 3.1 Flash-Lite Preview obeys the persona by between +0.574 and +0.752 at every one of the ten scales while Gemini 3.5 Flash-Lite runs from −0.489 to +0.291 over the same grid. Consistent with that, the three-way interaction between model, regime and persona is itself significant, at $F(4, \cdot) = 53.3$, $P = 1.3 \times 10^{-44}$ among the ten-scale models and $F(2, \cdot) = 32.7$, $P = 8.1 \times 10^{-15}$ among the three-scale ones, so the pooled two-way interaction is not a summary of a pattern the models share.

What the models do share is the direction of the shift. Measured as a difference in differences with its own dyad bootstrap, the persona effect moves toward compliance across the fractional-to-unit boundary in every one of the six, by +0.340, +0.088 and +0.090 among the ten-scale models and +0.256, +0.691 and +0.396 among the three-scale ones, and all six intervals exclude zero. Where the sweep continues into the large regime the further shift is small and not shared in sign, at −0.034, −0.033 and +0.018, none of the three excluding zero. The common movement is into the unit regime and not back out of it, which is a weaker statement than a coarser grid supported, and it is the one this corpus carries.

The two statements have to be kept apart, and separating them is what the result is. *Whether* a model obeys a cooperative persona at all is a property of that model in half of this corpus, stable across five orders of magnitude in the payoff scale. *How strongly* it obeys is not a property of the model: it is shifted, in a common direction and by as much as 0.69, by a change to the payoff matrix that carries no instruction, no information about the task, and no game-theoretic content whatever. For three of the six models that shift is large enough to carry the effect across zero, which is how the same system prompt comes to raise cooperation at one scale and lower it at another in the same model.

3.9 Language and objective framing

The prompt language interacts with the scale as well (figure 5b). Among the ten-scale models the spread between the five languages ranges from 0.055 to 0.228 depending on which scale it is measured at, and the scale swing ranges from 0.133 to 0.196 depending on which language it is measured in (electronic supplementary material, table S8); the interaction between the notation regime and language is $F(8, \cdot) = 10.3$, $P = 1.9 \times 10^{-14}$. The three-scale models give the same result at a smaller size, $F(4, \cdot) = 6.8$, $P = 1.9 \times 10^{-5}$. The languages do not move in parallel: cooperation in English rises from 0.641 at $\lambda = 0.1$ to 0.812 at $\lambda = 0.25$ and falls back to 0.662 at $\lambda = 1000$, while cooperation in Chinese runs 0.635, 0.648 and 0.650 over the same three scales and never leaves a band 0.14 wide across the whole sweep.

One feature of the FAIRGAME templates makes a partial mechanism test available, and

Table 1: The persona instruction, by model and notation regime. The effect is the cooperation rate of an agent told to be cooperative minus that of an agent told to be selfish, measured within a regime. "Shift" is the difference between two of those effects, with its own bootstrap over whole dyads. A dagger marks an entry whose 95% interval excludes zero. The last column records whether the effect keeps one sign over the whole sweep. Read down the shift columns: every model moves toward compliance entering the unit regime, which is the shared pattern; none of the three shifts out of it excludes zero. Read down the effect columns: three of the six never change sign, which is why the pooled curve of §3.8 describes no individual model. The same effects scale by scale rather than regime by regime are in the electronic supplementary material, table S4.

		Effect of the cooperative persona			Shift		
Sweep	Model	fractional	unit	large	frac.→unit	unit→large	Sign
ten scales	Gemini 3.5 Flash-Lite	$-0.099^{\dagger}$	$+0.241^{\dagger}$	$+0.207^{\dagger}$	$+0.340^{\dagger}$	-0.034	**crosses**
ten scales	Gemini 3.1 Flash-Lite Prev.	$+0.650^{\dagger}$	$+0.737^{\dagger}$	$+0.704^{\dagger}$	$+0.088^{\dagger}$	-0.033	constant
ten scales	GPT-5.4 Nano	$-0.072^{\dagger}$	$+0.019$	$+0.037$	$+0.090^{\dagger}$	$+0.018$	**crosses**
three scales	Claude 3.5 Haiku	$-0.318^{\dagger}$	$-0.062^{\dagger}$	-	$+0.256^{\dagger}$	-	constant
three scales	GPT-4o	$-0.438^{\dagger}$	$+0.252^{\dagger}$	-	$+0.691^{\dagger}$	-	**crosses**
three scales	Mistral Large	$-0.599^{\dagger}$	$-0.203^{\dagger}$	-	$+0.396^{\dagger}$	-	constant

The three-scale models have no large regime, because their sweep stops at $\lambda = 10$. Model × regime × persona interaction: $F(4, \cdot) = 53.3$, $P = 1.3 \times 10^{-44}$ (ten scales); $F(2, \cdot) = 32.7$, $P = 8.1 \times 10^{-15}$ (three scales).

we use it rather than treat it only as a limitation. The templates do not state the objective identically in every language: English, French and Vietnamese tell the agent to minimise a penalty, while Chinese and Arabic tell it to maximise a reward, with the cell values still described as penalties. Language is therefore confounded with objective framing in this corpus, which means the five-language contrast is an omnibus test and we report it as one.

The confound is informative here because a natural account of the scale effect predicts something specific about it. If models were reading the displayed numbers as rewards to be maximised rather than penalties to be minimised, and if raising λ made the largest attainable number easier to notice, then the direction of the scale effect should differ between the two framings. It does not. Across the ten scales the maximise-framed languages swing by 0.144 and the minimise-framed ones by 0.131, and neither is monotone; the framing itself shifts the level of cooperation by only 0.026 among the ten-scale models and 0.022 among the three-scale ones, an order of magnitude less than the language spread it is confounded with. What the two framings do not do is respond in opposite directions, which is what the reward-reading account requires. They do not respond identically either, and we say so: the interaction between the notation regime and the framing is significant at $F(2, \cdot) = 23.8$, $P = 5.0 \times 10^{-11}$. The scale effect is therefore not a reward-reading effect, or at least not only one.

3.10 The strategy label moves, and it moves selectively

Cooperation rate is one summary. A second, and the one that feeds evolutionary models of mixed populations, is the strategy a read-out assigns to a whole run of play. We pass all 15,600 agent-games through the hybrid read-out of §2.4, which proves a label from the four canonical memory-one rules where one of them exactly reproduces the run, records the run as ambiguous where several do, and otherwise takes the argmax of an LSTM trained on synthetic play from the same four. The mix over those five categories is the dependent variable (figure 5c).

It moves with the scale in all six models. The largest total-variation distance between the mixes at two scales is 0.338 for Gemini 3.5 Flash-Lite, 0.313 for GPT-5.4 Nano and 0.160 for Gemini 3.1 Flash-Lite Preview, against dyad-level permutation nulls of 0.155, 0.143 and 0.145; and 0.263 for Claude 3.5 Haiku, 0.325 for GPT-4o and 0.125 for Mistral Large, against nulls of 0.115, 0.103 and 0.098. Every one of the six exceeds its null, at $P < 0.001$ for four, $P = 0.006$ for Mistral Large and $P = 0.016$ for Gemini 3.1 Flash-Lite Preview. Restricting the vocabulary to the four named rules, so that movement into and out of the ambiguous category cannot contribute, leaves all six significant at $P \leq 0.004$, at 0.304, 0.320, 0.194, 0.245, 0.296 and 0.127 in the same order. The restriction raises three of the six estimates rather than lowering them, so none of these shifts is an artefact of play becoming more or less nameable.

The shape of the movement is the substantive part, and a single distance cannot show it (figure 5c, table 2). Resolved into one curve per label, the AllD share over the ten scales runs 0.254, 0.253, 0.162, 0.186, 0.176, 0.187, 0.174, 0.207, 0.187 and 0.213, and the AllC share runs 0.320, 0.312, 0.374, 0.345, 0.382, 0.360, 0.370, 0.351, 0.342 and 0.340: the pooled cooperation curve and its mirror image, with the same step out of the two smallest scales and the same absence of movement above them. Over the three scales the same two labels move much further, AllD falling from 0.440 to 0.236 and AllC rising from 0.302 to 0.372 between $\lambda = 0.1$ and $\lambda = 10$. The scale result therefore holds on a read-out that consumes the whole sequence, and not only on an average over rounds.

The two conditional rules move less, and by how much depends on the grid, which is worth stating carefully because a coarse grid overstates the contrast. Over the three scales the ordering is sharp: AllD moves 0.204 while TFT moves 0.034 and WSLS 0.056, a factor of six and of four. Over the ten scales it is much weaker: AllD moves 0.092 against 0.051 for TFT and 0.052 for WSLS, and all five labels including *ambiguous* clear their nulls ($P \leq 0.016$). What the payoff scale moves most is how often an agent is assigned an *unconditional* rule and which one; but the claim that reciprocal play is untouched does not survive ten scales, any more than the coordination invariant of §3.7 does, and the two failures are the same fact seen at the level of the sequence and of the round.

Two further observations belong here because they qualify what the read-out can be asked to support. First, Mistral Large is the model whose cooperation rate did not move, and its strategy mix does, at $P = 0.006$ over the five categories and $P = 0.004$ over the four named rules. A model can cooperate at the same rate under two scales and still be assigned a different distribution over strategies, because the rate is an average and the label is a claim about the sequence; invariance on one summary is not invariance on the other. Second, the read-out's own coverage is not constant across the sweep. The share of agent-games carrying a provable rule falls from 0.199 at $\lambda = 0.01$ to 0.142 at $\lambda = 0.5$ and recovers to 0.173 at $\lambda = 1000$ over the ten scales, and falls from 0.368 to 0.215 over the three, while the share sitting below the posterior floor rises from 0.211 to 0.302 over the three. Play is at its least exactly nameable at the scales where cooperation is highest, which is a property of the instrument as much as of the behaviour, and it is a further reason to read the named-vocabulary column of table 3 alongside the five-category one rather than instead of it.

Table 2: Each strategy label against the payoff scale. Shares are of all agent-games at that scale, within the group of models named on the block header. "Swing" is the largest difference between two scales, tested against a permutation null that shuffles the scale label across whole dyads, exactly as in table 3. The predicted swing is zero on every row. On the three-scale grid the two unconditional rules move by four to six times what the two conditional ones do; on the ten-scale grid every label moves, and the ratio falls below two. "Ambiguous" is not a strategy: it records that several rules reproduce the run of play equally well, so its row measures how nameable the play is rather than what was played.

	share of agent-games at $\lambda =$										across the sweep		
Label	0.01	0.1	0.25	0.5	1	2	5	10	100	1000	swing	null	P
the three models that met ten scales													
AllD	0.254	0.253	0.162	0.186	0.176	0.187	0.174	0.207	0.187	0.213	0.092	0.053	< 0.001
AllC	0.320	0.312	0.374	0.345	0.382	0.360	0.370	0.351	0.342	0.340	0.069	0.060	0.009
TFT	0.094	0.080	0.086	0.103	0.112	0.119	0.113	0.108	0.131	0.128	0.051	0.041	0.002
WSLS	0.119	0.139	0.133	0.137	0.172	0.138	0.153	0.147	0.142	0.138	0.052	0.047	0.016
Ambiguous	0.212	0.215	0.245	0.229	0.158	0.196	0.189	0.188	0.198	0.181	0.087	0.073	0.008
the three models that met three scales													
AllD	-	0.440	-	-	0.240	-	-	0.236	-	-	0.204	0.047	< 0.001
AllC	-	0.302	-	-	0.386	-	-	0.372	-	-	0.084	0.043	< 0.001
TFT	-	0.105	-	-	0.136	-	-	0.139	-	-	0.034	0.032	0.035
WSLS	-	0.142	-	-	0.188	-	-	0.198	-	-	0.056	0.039	0.004
Ambiguous	-	0.012	-	-	0.050	-	-	0.055	-	-	0.043	0.027	< 0.001

Provenance of the labels (§2.4), as a share of all agent-games in the group: provable rule 17.2% over ten scales and 26.7% over three, ambiguous 20.1% and 3.9%, LSTM above the 0.90 posterior floor 38.7% and 43.5%, LSTM below it 24.0% and 25.9%. None of these four shares is constant across the sweep; the breakdown by scale is in the electronic supplementary material, table S7, and the mix resolved by model in table S6.

4 Discussion

4.1 A cooperation rate is a statement about one choice of units

The central result is that the strategic behaviour of a language model is not invariant to the one manipulation whose null is a theorem. Five of six models move by up to 0.24 in cooperation rate, and all six move by up to 0.34 in total variation over strategy labels, when every payoff is multiplied by a positive constant. This is not a framing effect in the sense the term usually carries, because no description changed. The words of the prompt, the ordering of the options, the identity of the dominant action and the structure of the dilemma are all identical between two conditions that differ by a factor of ten.

The consequence for measurement is that a published cooperation rate for a language model is conditional on a design choice that the theory says is not a choice, and that is therefore almost never reported. The size of that conditioning is not small. Among the three models that swept ten scales, the largest within-model swing exceeds the median spread between the three at a fixed scale, and 24 of 135 model-pair comparisons change sign somewhere in the sweep. Two of the three models on the narrower grid change places between $\lambda = 0.1$ and $\lambda = 1$. A field that compares models on cooperation is comparing them at whichever scale each paper happened to write down.

Table 3: Scale sensitivity per model. "Swing" is the largest difference in cooperation rate between two payoff scales and "at" names the scale where cooperation peaks. "Monotone" records whether the response increases or decreases throughout the sweep. "Coop. shift" repeats the swing as a test against a permutation null that shuffles the scale label across whole dyads, and "mix shift" is the same test on the largest total-variation distance between the strategy mixes at two scales, reported over the five categories the read-out emits and again over the four named rules alone, on which movement into the ambiguous category cannot contribute. "Best description" is the specification of table 4 with the lowest BIC for that model. The predicted value of every shift column is exactly zero, because a positive rescaling of a von Neumann–Morgenstern utility leaves the game unchanged. The two groups of models differ in whether the round count was disclosed in the prompt and in sampling temperature (§2.6), so no level statistic is comparable between them; every shift is measured within a model.

			Cooperation			Coop. shift		Mix shift		
Sweep	Model	Coop.	swing	at	Mono.	obs.	null	5-cat.	named	null
ten scales	Gemini 3.5 Flash-Lite	0.751	0.190	λ=0.25	no	0.190^{***}	0.082	0.338^{***}	0.304^{***}	0.155
ten scales	Gemini 3.1 Flash-Lite Prev.	0.617	0.072	λ=0.1	no	0.072	0.077	0.160^{*}	0.194^{***}	0.145
ten scales	GPT-5.4 Nano	0.606	0.243	λ=5	no	0.243^{***}	0.080	0.313^{***}	0.320^{***}	0.143
three scales	Claude 3.5 Haiku	0.625	0.240	λ=10	yes	0.240^{***}	0.058	0.263^{***}	0.245^{***}	0.115
three scales	GPT-4o	0.448	0.234	λ=10	yes	0.234^{***}	0.062	0.325^{***}	0.296^{***}	0.103
three scales	Mistral Large	0.524	0.036	λ=1	no	0.036	0.065	0.125^{**}	0.127^{**}	0.098

$^{***}P < 0.001$, $^{**}P < 0.01$, $^{*}P < 0.05$, blank $P \geq 0.05$, over 2,000 dyad-level permutations; the null column is the 95th percentile of the five-category null, and the named-vocabulary nulls, which run from 0.094 to 0.150, are given in the electronic supplementary material, table S5. Best description by BIC: the specification saturated in λ for Gemini 3.5 Flash-Lite, a linear term in $\log_{10} \lambda$ for Gemini 3.1 Flash-Lite Preview, a quadratic for GPT-5.4 Nano, the fractional flag for Claude 3.5 Haiku and GPT-4o, and no λ term at all for Mistral Large, which is the correct description of a model whose cooperation rate did not move.

4.2 Notation, and what the account can and cannot claim

An earlier version of this analysis reported that the response is a step function over how the matrix is written. On the three-scale grid that conclusion is what the numbers say, and we no longer hold it. It does not survive the denser sweep, and the way it fails is instructive enough to state plainly rather than to quietly drop.

On three scales the notation description wins by 8.1 BIC points over the specification saturated in λ. On three scales those two specifications also fit identically, to five decimal places in r^2, and the advantage is exactly the penalty BIC charges for one extra parameter at that sample size. The winner was not a better account of the data. It was the same account of the data written with one parameter fewer, on a grid with too few points for any account to be distinguished from any other. Three scales cannot separate a quadratic from a cubic from a saturated model either: all three have the same BIC to six decimal places, because with three points they are the same model. A ladder of specifications is only informative when the rungs are distinguishable, and nothing in the fit statistics announces when they are not.

With ten scales they are distinguishable, and no notation specification wins in any model. The regime read from the printed cells, which is the operational form the notation account actually requires, does worse than the regime read from λ in both models where the two differ. That is the diagnostic result: if the account were about how the numbers look, the version built from how the numbers look should be the better one.

The experiment that separates the accounts is no longer a proposal. Holding λ fixed and

Table 4: The description ladder, pooled within each group. Every specification predicts the cooperation rate of an agent-game from the same controls for model, prompt language, own persona and opponent persona, and differs only in how the payoff scale enters. ΔBIC is measured from the best specification within each group; k counts the free parameters the specification adds to the controls. The saturated row is the ceiling on what any function of λ can fit, so a row that matches it at lower k has found the shape of the response. On three scales the fractional flag, the regime factor, the quadratic, the cubic, the glyph count and the saturated specification collapse onto two distinct fits, and the 8.1 points separating them is the penalty for one parameter at $n = 3{,}600$, which is $\ln n = 8.19$. Over three scales a cubic is collinear with a quadratic and only two notation regimes are present, so the fractional flag and the regime factor are the same specification. The pooled ten-scale column is reported for symmetry and should be read against figure 3b: the three models it averages prefer three different specifications, so no row of it describes any of them.

	ten scales, pooled		three scales, pooled	
How λ enters	k	ΔBIC	k	ΔBIC
not at all	0	8.4	0	191.9
linear in $\log_{10}\lambda$	1	17.8	1	47.9
quadratic in $\log_{10}\lambda$	2	**0.0**	2	8.1
cubic in $\log_{10}\lambda$	3	5.9	-	8.1
fractional flag	1	17.0	1	**0.0**
fractional flag + glyph count	2	18.9	2	8.1
notation regime, 3 levels	2	24.3	1	**0.0**
saturated in λ	9	14.4	2	8.1

rendering the same payoffs as decimals raises cooperation by about 0.06 at every scale in one of the two models tested, against a placebo rendering change that moves 0.02, and does nothing interpretable in the other (§3.3). So notation can matter, causally, in a model. What it produces is a level shift that is the same size at $\lambda = 0.1$ and at $\lambda = 10$, and a constant offset cannot generate a dependence on λ. This is why the same model can show a real notation effect and still rank the notation description 11.0 BIC points below a linear term in $\log_{10}\lambda$: the two statements are about different things, one about the level and one about the shape.

The dependence on scale itself is not explained away by any of this. Holding the rendering fixed, so that every cell prints with the same kind of glyph at every scale, cooperation still moves by 0.09 across a single decade (§3.3). Whatever the response is a function of, it is a function of something that survives fixing the notation, and the invariance is violated either way.

4.3 An instruction is only as good as the numbers next to it

The persona result is the one we would put first in front of anyone building agent systems, and it needs stating in two parts because the two carry different weight.

The part that is common to every model is a shift. The strength with which a model follows a cooperative persona moves toward compliance when the payoffs next to that instruction are written as short integers rather than decimals, in all six models, by between 0.09 and 0.69. Nothing about the instruction changed, nothing about the task changed, and nothing a game theorist can compute from the matrix changed. Instruction-following is normally treated as a property of a model, measured and reported as such; what this says is that it is a property of a model and its numeric context jointly, and that the relevant part of the context can be a part carrying no instruction and no information about the task beyond its units. An evaluation of whether a model follows a behavioural instruction, run at one payoff scale, does not transfer to another scale of the same task.

The part that is *not* common to every model is the sign, and we were careful in §3.8 not to claim otherwise, because the pooled average invites it. Three of the six models obey a cooperative persona at one payoff scale and invert it at another; the other three keep whatever disposition they have across five orders of magnitude. Whether a system prompt is followed or reversed therefore remains a model-level fact worth measuring once. How much it is followed is not, and cannot be reported without the scale it was measured at. A practitioner should read the first as a property to select a model on, and the second as a reason to test the prompt at the units the deployment will actually use.

4.4 Consequences for models of hybrid populations

Work that models populations in which artificial and human agents interact [2, 13, 14], and the evolutionary-game machinery it draws on [24, 25], assigns agents strategies from a known family and computes the resulting dynamics. Three of our results bear on that programme directly. First, the assignment is not stable under a transformation the dynamics cannot see: the replicator equation is identical at λ and 10λ, and the strategy mix we would feed into it differs by up to 0.34 in total variation, or 0.32 over the four named rules alone. Two population models that a game theorist would call the same model therefore predict differently about the same agent.

Second, the instability sits in exactly the coordinate such models are most sensitive to. What moves most is the split between AllC and AllD, by 0.07 and 0.09 over the ten scales and by 0.08 and 0.20 over the three, and the seeded proportion of unconditional cooperators and defectors is what sets whether a replicator trajectory runs to fixation or to coexistence. Third, and cutting the other way, the reciprocal strategies move least: TFT by 0.05 and WSLS by 0.05 over the ten scales, against AllD's 0.09, and by 0.03 and 0.06 against 0.20 over the three. That is a difference of degree and not of kind, and we do not overstate it: on the denser grid the reciprocal labels move too, and clear their nulls. Since it is the reciprocal share that governs whether cooperation can be sustained against invasion at all [3, 20], the part of the mix that carries the qualitative behaviour of the dynamics is the part the rescaling disturbs least, and the part it moves most is the initial condition. That is a narrower problem than the raw total-variation number suggests, and a more tractable one.

The practical implication is not that the programme is unworkable. It is that a population model built on measured LLM strategies inherits a free parameter that the theory does not contain, that the parameter enters through the initial mix rather than through the reciprocal core, and that it should be swept rather than fixed.

4.5 Relation to the prompt-sensitivity and framing literatures

That language model outputs move with features of the prompt that carry no task content is established: formatting choices, separators and option order all shift accuracy [26, 27], and personas shift behaviour in ways that outrun what the persona states [28, 29]. Our contribution is not that a spurious feature matters, but that the feature here is one the relevant theory has already declared to be spurious with a proof. In an accuracy benchmark, the claim that a separator should not matter is an intuition. In a game, the claim that units should not matter is the representation theorem, and that is what makes the size of the failure interpretable rather than merely surprising.

The relation to the human framing literature runs the other way [30, 31]. Human framing effects are produced by re-describing the same outcomes, and they map to a compact set of psychological mechanisms: reference points, loss aversion, the salience of a gain or a loss. Nothing is re-described here. Within an arm the two conditions differ by a multiplicative constant, which is the one change that prospect theory and expected utility theory agree in treating as inert once the units of the value function are fixed. Our closest neighbour in the LLM literature is the finding of Lorè and Heydari [12] that contextual framing carries weight comparable to game structure; we document a sensitivity one step further out, to a change that is not a framing at all.

4.6 Limitations

Six limitations bound these conclusions.

Three of the six models met only three scales, and they contribute nothing to any claim about the shape of the response. We showed in §3.2 that a three-point grid cannot separate the specifications it appears to rank, so those models enter only through results that need no resolution in λ. Every statement about shape rests on three models, and two of the three are from one provider.

The rendering experiment of §3.3 was collected as separate runs from the main sweep, so a run-level offset is confounded with the rendering change by construction. The placebo condition is what bounds that offset, and it bounds it at about 0.02; the effect we interpret is 0.06. A within-run randomisation of the rendering would remove the confound rather than bound it, and is what we would do next.

We test the multiplicative half of the affine invariance and not the additive half. Adding a constant to every cell is the weaker of the two demands, since it does not change the number of digits, and we would expect a smaller effect; but it has a cleaner interpretation, because it moves the reference point without touching the notation, and it is the natural companion experiment. We note one complication that the present design makes visible: because the templates state payoffs as penalties to be minimised, adding a constant in utility space is subtracting one in the printed space, so the additive experiment is not simply the multiplicative one with a different operator.

The sweep is dense enough to rank specifications and not dense enough to locate features. Ten scales over five orders of magnitude place the turning points of the two non-monotone models only to within the gaps between sampled scales, and the three-level regime partition remains a coarse summary. The claim that a smooth function of $\log_{10} \lambda$ is the wrong shape in one model, and the right shape in another, does not depend on the spacing; any claim about where exactly a curve turns does.

The strategy result rests on an instrument, and the instrument is imperfect in a way that is worth naming rather than absorbing. Two thirds of the labels are supplied by a learned classifier rather than by a proof, and a quarter of all agent-games sit below the posterior floor at which a read-out designed to abstain would decline to answer (§2.4). We have reported every test on the four named rules as well as on all five categories, and the conclusions of §3.10 hold on both; but the exact-rule coverage of this corpus is lower than the classifier's training distribution, so its labels are nearest neighbours more often than they are recognitions. What would remove the caveat is an abstaining read-out, reported with its coverage, rather than a larger vocabulary.

Finally, the model set is six frontier models, none of them with an explicit reasoning procedure, and reasoning traces were disabled by design so that the parser could not manufacture cooperation. That exclusion is not incidental: a model that emits hidden reasoning exhausts a small output budget inside the trace and returns a truncated string, which a harness then resolves to a default action, and the resulting cooperation rate is the parser's rather than the model's. We report parse-failure and fallback rates of zero for every cell in this corpus precisely because reasoning was off. Whether a model that reasons at inference time recovers the invariance is the single most valuable extension, and it is the one experiment whose result we would not want to predict.

4.7 A reporting standard, and what it costs

The remedy is cheap, and it follows from the first result alone. Any measurement of strategic behaviour in a language model should report the scale at which the payoff matrix was written, and should report the measurement at more than one scale. Two conditions one decade apart are enough to detect the effect, but *which* decade is not a detail, and the corpus is specific about it (electronic supplementary material, table S9). Held to the null of each model's full sweep, which is conservative because a two-level design generates a smaller one, the contrast $\lambda = 0.1$ against $\lambda = 1$ detects the effect in the two three-scale models that move at all, at gaps of 0.232 and 0.226 against nulls of 0.058 and 0.062, and in none of the three models that swept ten scales, where the same contrast gives 0.035, 0.013 and 0.080 against nulls of 0.080, 0.077 and 0.082. Moving the pair one decade down, to $\lambda = 0.01$ against $\lambda = 0.1$, detects it in two of those three, at 0.103 and 0.098; so does the off-decade pair $\lambda = 0.5$ against $\lambda = 5$, at 0.082 and 0.104. The informative decade on this corpus is the small-stakes end, which is the end no published design samples. Three conditions spanning two decades additionally recover the shape. The cost is a factor of two or three in sampling on a quantity that is otherwise reported as though it were a constant of the model.

Concretely, we propose that a cooperation rate be published alongside its *scale swing*, the largest difference between two scales, and its permutation null. On this corpus those pairs are 0.190 against 0.082, 0.072 against 0.077, 0.243 against 0.080, 0.240 against 0.058, 0.234 against 0.062 and 0.036 against 0.065. Four of the six exceed the null, and the two that do not are the two discussed in §3.1: one is a genuine negative and the other is a shape against which a largest-gap statistic has almost no power. Neither is expressible in the current practice of reporting one number.

4.8 Conclusion

Six frontier language models played 7,800 iterated prisoner's dilemmas in which the only manipulation of the game was to multiply every payoff by a positive constant, a transformation whose effect on any theory of the game is exactly zero. Cooperation moved in five of the six, by up to 0.24, and the strategy mix moved in all six. The response is not monotone in the payoff magnitude, and no two of the three models dense enough to show a shape have the same one, so pooling across models reports a large effect as a small one. A description built from how the numbers are printed wins on the three-scale grid that most designs sample, by exactly the penalty for one parameter at equal fit, and loses in every model once ten scales are available. The effect is already open on the opening move, so it is a prior the prompt's numbers induce and not something learned from play. It shifts how far a persona instruction is obeyed, in the same direction in every model, and it interacts with the prompt language. What it disturbs least is reciprocity: how well the two agents coordinate, and how often either is assigned a rule that responds to the other, move less than anything else we measured, though on the densest grid they move too.

The reading we draw is not that these models fail to be rational agents, which would be a small claim about a well-known fact. It is that the quantity a benchmark reports is jointly determined by the game and by an arbitrary choice of units that the theory promised was free. Until that choice is swept rather than fixed, a cooperation rate attributed to a model is attributed too widely.

Ethics

This work involved no human participants and no animal subjects, and required no ethical approval. The agents studied are commercial language models, queried through their providers' interfaces under the respective terms of service. No personal data were collected, generated or processed at any stage: every prompt is a payoff matrix, a persona sentence and a history of prior moves, and every response is a single option token.

Data accessibility

The full corpus of 7,800 dyads and 156,000 logged decisions in the main sweep, the 4,000 further dyads of the rendering and stag-hunt experiments, the open-weight corpus of the earlier preprint reported in the electronic supplementary material, the analysis pipeline that produces every figure and table in this paper and in that document, and the prompt templates are available at [REPOSITORY URL]. Every number appearing in a figure or a table is also written to a machine-readable CSV file in the same repository, and each figure and supplementary table is generated from those files rather than transcribed, so no displayed value can drift from the analysis that produced it. A permanent archive with a DOI will be deposited on acceptance.

Additional material is deposited with the manuscript as electronic supplementary material: §S1 there validates the strategy read-out, §S2 reports the open-weight corpus of the earlier preprint on which no claim in this paper rests, and tables S1–S10 give the per-model and per-cell breakdowns behind the summary statistics reported here.

Declaration of AI use

Large language models are the object of study of this paper rather than a tool used to produce it: the corpus consists of their logged behaviour, and the models, versions and decoding settings are stated in §2.2. [AUTHORS TO COMPLETE: declare here any use of generative AI in preparing the manuscript itself, for example in drafting, editing or code generation, in accordance with the journal's policy.]

Authors' contributions

[A.O.]: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing - original draft; [A.T.]: conceptualization, methodology, validation, supervision, funding acquisition, writing - review and editing.

Both authors gave final approval for publication and agreed to be held accountable for the work performed herein.

Conflict of interest declaration

The authors declare no competing interests.

Funding

[FUNDING STATEMENT]

Acknowledgements

We thank the FAIRGAME developers at the Luxembourg Institute of Science and Technology for releasing the multilingual game templates on which the language manipulation is built.

References

[1] Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, 2023. doi: 10.1145/3586183.3606763.

[2] Allan Dafoe, Yoram Bachrach, Gillian Hadfield, Eric Horvitz, Kate Larson, and Thore Graepel. Cooperative AI: machines must learn to find common ground. *Nature*, 593:33–36, 2021. doi: 10.1038/d41586-021-01170-0.

[3] Robert Axelrod and William D. Hamilton. The evolution of cooperation. *Science*, 211 (4489):1390–1396, 1981. doi: 10.1126/science.7466396.

[4] Robert Axelrod. *The Evolution of Cooperation*. Basic Books, New York, 1984.

[5] Martin A. Nowak and Karl Sigmund. Tit for tat in heterogeneous populations. *Nature*, 355: 250–253, 1992. doi: 10.1038/355250a0.

[6] Martin A. Nowak. Five rules for the evolution of cooperation. *Science*, 314(5805):1560–1563, 2006. doi: 10.1126/science.1133755.

[7] William H. Press and Freeman J. Dyson. Iterated Prisoner's Dilemma contains strategies that dominate any evolutionary opponent. *Proceedings of the National Academy of Sciences*, 109(26):10409–10413, 2012. doi: 10.1073/pnas.1206569109.

[8] Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. Playing repeated games with large language models. *Nature Human Behaviour*, 9: 1380–1390, 2025. doi: 10.1038/s41562-025-02172-y.

[9] Steve Phelps and Yvan I. Russell. Investigating emergent goal-like behaviour in large language models using experimental economics. *arXiv preprint arXiv:2305.07970*, 2023.

[10] Nicolò Fontana, Francesco Pierri, and Luca Maria Aiello. Nicer than humans: How do large language models behave in the prisoner's dilemma? In *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)*, pages 522–535, 2025. doi: 10.1609/icwsm.v19i1.35829.

[11] Qiaozhu Mei, Yutong Xie, Walter Yuan, and Matthew O. Jackson. A Turing test of whether AI chatbots are behaviorally similar to humans. *Proceedings of the National Academy of Sciences*, 121(9):e2313925121, 2024. doi: 10.1073/pnas.2313925121.

[12] Nunzio Lorè and Babak Heydari. Strategic behavior of large language models and the role of game structure versus contextual framing. *Scientific Reports*, 14:18490, 2024. doi: 10.1038/s41598-024-69032-z.

[13] Ana Paiva, Fernando P. Santos, and Francisco C. Santos. Engineering pro-sociality with autonomous agents. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32, 2018. doi: 10.1609/aaai.v32i1.12215.

[14] Fernando P. Santos, Francisco C. Santos, and Jorge M. Pacheco. Social norm complexity and past reputations in the evolution of cooperation. *Nature*, 555:242–245, 2018. doi: 10.1038/nature25763.

[15] Iyad Rahwan, Manuel Cebrian, Nick Obradovich, Josh Bongard, Jean-François Bonnefon, Cynthia Breazeal, Jacob W. Crandall, Nicholas A. Christakis, Iain D. Couzin, Matthew O. Jackson, Nicholas R. Jennings, Ece Kamar, Isabel M. Kloumann, Hugo Larochelle, David Lazer, Richard McElreath, Alan Mislove, David C. Parkes, Alex Pentland, Margaret E. Roberts, Azim Shariff, Joshua B. Tenenbaum, and Michael Wellman. Machine behaviour. *Nature*, 568:477–486, 2019. doi: 10.1038/s41586-019-1138-y.

[16] John von Neumann and Oskar Morgenstern. *Theory of Games and Economic Behavior*. Princeton University Press, Princeton, NJ, 1944.

[17] R. Duncan Luce and Howard Raiffa. *Games and Decisions: Introduction and Critical Survey*. Wiley, New York, 1957.

[18] Author One and Author Two. Payoff scaling shapes cooperation in LLM agents across languages, 2026. Preprint by the present authors.

[19] Alessio Buscemi, Daniele Proverbio, Alessandro Di Stefano, The Anh Han, German Castignani, and Pietro Liò. FAIRGAME: A framework for AI agents bias recognition using game theory. In *ECAI 2025 — Proceedings of the 28th European Conference on Artificial Intelligence*, pages 4097–4104. IOS Press, 2025. doi: 10.3233/FAIA251300.

[20] Martin A. Nowak and Karl Sigmund. A strategy of win-stay, lose-shift that outperforms tit-for-tat in the Prisoner's Dilemma game. *Nature*, 364:56–58, 1993. doi: 10.1038/364056a0.

[21] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997. doi: 10.1162/neco.1997.9.8.1735.

[22] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, New York, 1993.

[23] Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2): 461–464, 1978. doi: 10.1214/aos/1176344136.

[24] György Szabó and Gábor Fáth. Evolutionary games on graphs. *Physics Reports*, 446(4–6): 97–216, 2007. doi: 10.1016/j.physrep.2007.04.004.

[25] Matjaž Perc, Jillian J. Jordan, David G. Rand, Zhen Wang, Stefano Boccaletti, and Attila Szolnoki. Statistical physics of human cooperation. *Physics Reports*, 687:1–51, 2017. doi: 10.1016/j.physrep.2017.05.004.

[26] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design, or: How i learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*, 2024.

[27] Yao Lu, Max Bartolo, Alastair Moore, Sebastian Riedel, and Pontus Stenetorp. Fantastically ordered prompts and where to find them: Overcoming few-shot prompt order sensitivity. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 8086–8098, 2022. doi: 10.18653/v1/2022.acl-long.556.

[28] Ameet Deshpande, Vishvak Murahari, Tanmay Rajpurohit, Ashwin Kalyan, and Karthik Narasimhan. Toxicity in ChatGPT: Analyzing persona-assigned language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 1236–1270, 2023. doi: 10.18653/v1/2023.findings-emnlp.88.

[29] Leonard Salewski, Stephan Alaniz, Isabel Rio-Torto, Eric Schulz, and Zeynep Akata. In-context impersonation reveals large language models' strengths and biases. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

[30] Amos Tversky and Daniel Kahneman. The framing of decisions and the psychology of choice. *Science*, 211(4481):453–458, 1981. doi: 10.1126/science.7455683.

[31] Daniel Kahneman and Amos Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979. doi: 10.2307/1914185.